

LPO 8800: Statistical Methods in Education Research
Vanderbilt University
Midterm Exam
October 7, 2021

Name: _____

By signing below, I agree to the terms of Vanderbilt University's honor code.

Signature: _____

Instructions: Read each question carefully and provide clear, concise responses in the space provided. Be sure to complete every part of every question. Partial credit will be given where appropriate. If you make any assumptions to answer a question, please state those explicitly. You may use a calculator and one sheet of formulas/notes. Good luck!

Question 1 The following questions relate to the Early Childhood Longitudinal Study–Kindergarten Cohort (ECLS-K), a longitudinal study of more than 10,000 children followed from kindergarten through 8th grade. **(20 points)**

(a) Several variables from this dataset are described below. For each, indicate whether the variable is a categorical/qualitative or quantitative measure, and if quantitative, discrete or continuous. In addition, indicate for each whether you believe the variable’s measurement is nominal, ordinal, interval, or ratio. **(5 points)**

- Mother’s age in years

- Index of family’s socioeconomic status

- Census region (Northeast, Midwest, South, West)

- Reading scale score, 1st grade

- Teacher rating: how often the child works to the best of their ability (never, seldom, usually, always)

- (b) Use the following Stata output to provide the statistics requested below. The variable here is the student's reading test score in 1st grade. **(6 points)**

```
. summ C1R4RSCL if C1R4RSCL>0, detail
```

C1 RC4 READING IRT SCALE SCORE				

	Percentiles	Smallest		
1%	23.38	22.34		
5%	24.81	22.84		
10%	25.71	23.06	Obs	330
25%	30	23.38	Sum of Wgt.	330
50%	34.015		Mean	35.11512
		Largest	Std. Dev.	8.378345
75%	39	63.89		
90%	45.03	74.8	Variance	[omitted]
95%	49.04	77.76	Skewness	[omitted]
99%	63.89	85.91	Kurtosis	9.666257

Find the following:

- Inter-quartile range
- Variance
- Median
- 95% confidence interval for the population mean

- (c) Based on the information provided in part (b), how would you describe the shape/symmetry of the reading test score distribution? Briefly explain how you arrived at your answer. (**4 points**)
- (d) Your research assistant transformed the variable in part (b) to z -scores and reported back to you the average z -score for select subgroups of students. For example, students in higher-income households had a mean reading z -score of 0.85. How would you explain this value in words, and what is the mean reading score for students in higher-income households on the original scale? (**5 points**)

Question 2. Listed below are the joint probabilities that a randomly drawn middle school student has a computer at home (or not) and took an advanced placement (AP) course in high school (or not). For example, the probability a randomly drawn student has a computer at home and takes an advanced placement course in high school is 0.290. Use this information to answer the following questions. **(10 points)**

		Computer at home		Total
		No	Yes	
AP course in HS	No	0.235	0.185	
	Yes	0.290	0.290	
Total				1.000

(a) What is $P(\text{AP course})$? **(2 points)**

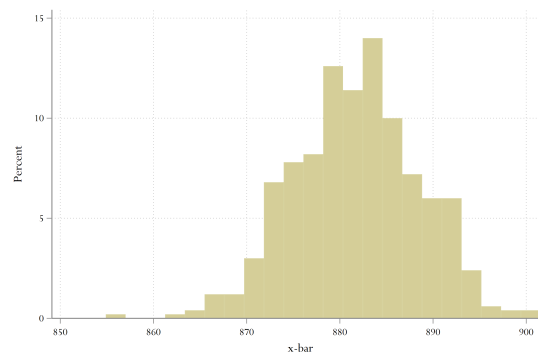
(b) What is $P(\text{AP course} \mid \text{computer at home})$? **(3 points)**

(c) Using the results above, determine whether taking an AP course in high school is independent of having a computer at home, and explain your answer. **(5 points)**

Question 3. The descriptive statistics and histogram below represent a simulated sampling distribution for $\bar{x}$ in which $n = 25$ for each sample from the population. 500 random samples were drawn as part of the simulation. **(10 points)**

```
. summ mean
```

Variable	Obs	Mean	Std. Dev.	Min	Max
-----+-----					
mean	500	881.7864	6.734863	854.8868	901.5418



(a) What is the (simulated) standard error of $\bar{x}$? Briefly explain your answer. **(5 points)**

(b) It is easy to show that the sample mean $\bar{x}$ is *unbiased*. Briefly explain what this means. **(5 points)**

Question 4. Student loan debt for graduates of Borrow U is believed to be normally distributed with a mean (μ) of \$18,400 and a standard deviation (σ) of \$4,700. Use this information to answer the following questions. **(25 points)**

- (a) Given the information provided above, approximately what fraction of graduates have more than \$25,000 in student loan debt? Provide a numeric answer, and indicate how you arrived at this answer. **(5 points)**

- (b) The institutional research office at Borrow U plans to take a random sample of 30 graduates and calculate their mean student loan debt ($\bar{x}$). What would you expect the sampling distribution of $\bar{x}$ to look like? That is, what would be its expected value, standard error, and distributional shape (if this can be determined)? Briefly explain your answer. **(5 points)**

- (c) Given your sample of students, you calculate a mean student loan debt of \$16,500. Assuming the information above about the population distribution is true, what is the probability of drawing a random sample of $n = 30$ with a mean of \$16,500 or less? Briefly justify your answer. **(5 points)**

(d) If the standard deviation in the population (σ) were \$5,500 rather than \$4,700, would your sample mean of \$16,500 be more or less likely than in part (c)? Briefly explain your answer. (**5 points**)

(e) If your sample were of 40 students rather than 30 (and the standard deviation remained at $\sigma = \$4,700$), would your sample mean of \$16,500 be more or less likely than in part (c)? Briefly explain your answer. (**5 points**)

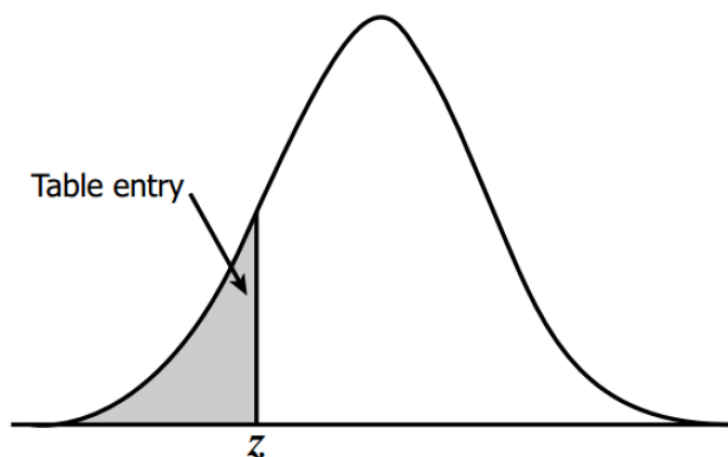
Question 5. In a recent Fox News poll of 512 Republican voters selected at random, 63% reported they think Donald Trump should run for president in 2024. Provide a 99% confidence interval for the true population proportion of Republican voters who think Donald Trump should run for president in 2024. Show your calculation. What is the margin of error? **(10 points)**

Question 6. Suppose you are constructing interval estimates for the population mean (and σ is unknown). In three cases below, I identify the confidence level and the sample size for three different sample means. For each case, identify the appropriate t value you would use in constructing the confidence interval. **(6 points—2 each)**

(a) 90% confidence interval and $n = 26$

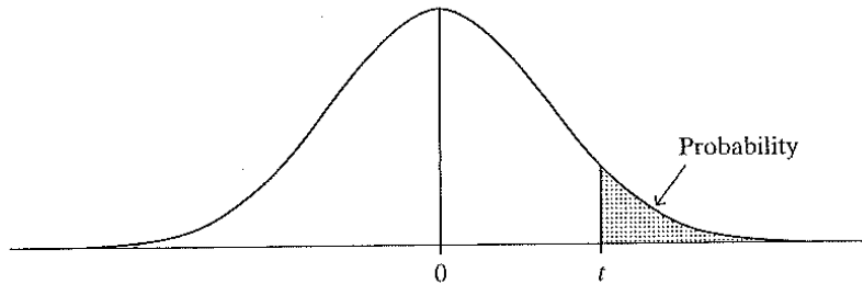
(b) 99% confidence interval and $n = 12$

(c) 80% confidence interval and $n = 20$



z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

TABLE B: t Distribution Critical Values



df	Confidence Level					
	80%	90%	95%	98%	99%	99.8%
	Right-Tail Probability					
	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	$t_{.001}$
1	3.078	6.314	12.706	31.821	63.656	318.289
2	1.886	2.920	4.303	6.965	9.925	22.328
3	1.638	2.353	3.182	4.541	5.841	10.214
4	1.533	2.132	2.776	3.747	4.604	7.173
5	1.476	2.015	2.571	3.365	4.032	5.894
6	1.440	1.943	2.447	3.143	3.707	5.208
7	1.415	1.895	2.365	2.998	3.499	4.785
8	1.397	1.860	2.306	2.896	3.355	4.501
9	1.383	1.833	2.262	2.821	3.250	4.297
10	1.372	1.812	2.228	2.764	3.169	4.144
11	1.363	1.796	2.201	2.718	3.106	4.025
12	1.356	1.782	2.179	2.681	3.055	3.930
13	1.350	1.771	2.160	2.650	3.012	3.852
14	1.345	1.761	2.145	2.624	2.977	3.787
15	1.341	1.753	2.131	2.602	2.947	3.733
16	1.337	1.746	2.120	2.583	2.921	3.686
17	1.333	1.740	2.110	2.567	2.898	3.646
18	1.330	1.734	2.101	2.552	2.878	3.611
19	1.328	1.729	2.093	2.539	2.861	3.579
20	1.325	1.725	2.086	2.528	2.845	3.552
21	1.323	1.721	2.080	2.518	2.831	3.527
22	1.321	1.717	2.074	2.508	2.819	3.505
23	1.319	1.714	2.069	2.500	2.807	3.485
24	1.318	1.711	2.064	2.492	2.797	3.467
25	1.316	1.708	2.060	2.485	2.787	3.450
26	1.315	1.706	2.056	2.479	2.779	3.435
27	1.314	1.703	2.052	2.473	2.771	3.421
28	1.313	1.701	2.048	2.467	2.763	3.408
29	1.311	1.699	2.045	2.462	2.756	3.396
30	1.310	1.697	2.042	2.457	2.750	3.385
40	1.303	1.684	2.021	2.423	2.704	3.307
50	1.299	1.676	2.009	2.403	2.678	3.261
60	1.296	1.671	2.000	2.390	2.660	3.232
80	1.292	1.664	1.990	2.374	2.639	3.195